

CLASS 9th
CHEMISTRY ADVANCED SHEET

A. Atomic Structure

1. Draw Bohr's model for ${}^{11}_5\text{B}$
2. If an element has 17 protons and 18 neutrons, write its symbol and electron configuration

B. Chemical Reactions

1. Classify and balance:
 - $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
 - $2\text{Al} + 3\text{CuCl}_2 \rightarrow 2\text{AlCl}_3 + 3\text{Cu}$
2. Identify oxidizing & reducing agents in: $\text{Zn} + \text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$

C. Periodic Table & Bonding

1. Compare reactivity of Na and K
2. Identify type of bond in: HCl, NaF, Mg

D. Acids, Bases & Salts

1. Find formula of salt formed from:
 - $\text{H}_2\text{SO}_4 + \text{NaOH} \rightarrow ?$
2. Name salt in reaction: $\text{AgNO}_3 + \text{NaCl} \rightarrow ?$

E. Metals & Non-Metals

1. Write reaction of Al with steam
2. Explain why metals are good conductors of electricity